

Q1: $f(x) = \frac{1}{\sqrt{(2\pi)^k |\Sigma|}} e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)}$, $x, \mu \in \mathbb{R}^k$, Σ 為 $k \times k$ 正定矩陣

說明 $\int_{\mathbb{R}^k} f(x) dx = 1$

pf: 1. $\because \Sigma$ 為正定, 由 Cholesky decomposition

$\longrightarrow \Sigma = LL^T$ L 為下三角矩陣

$\det(\Sigma) = \det(L) \det(L^T) = \det(L)^2$

$\det(L) = \sqrt{\det(\Sigma)}$

2. 以仿射轉換: $y = L^{-1}(x - \mu) \iff x = \mu + Ly$

$\longrightarrow dx = |\det(L)| dy$

$\|y\|^2 = y^T y = (L^{-1}(x - \mu))^T L^{-1}(x - \mu)$

$= (x - \mu)^T (L^{-T} L^{-1})(x - \mu)$

$= (x - \mu)^T (LL^T)^{-1}(x - \mu) = (x - \mu)^T \Sigma^{-1}(x - \mu)$

3. 代 $\lambda f(x) = \int_{\mathbb{R}^k} f(x) dx = \int_{\mathbb{R}^k} \frac{1}{\sqrt{(2\pi)^k |\Sigma|}} e^{-\frac{1}{2}(x-\mu)^T \Sigma^{-1}(x-\mu)} dx$

$= \frac{1}{\sqrt{(2\pi)^k |\Sigma|}} \int_{\mathbb{R}^k} \exp(-\frac{1}{2}\|y\|^2) |\det(L)| dy$

$= \frac{1}{\sqrt{(2\pi)^k |\Sigma|}} \sqrt{\det(\Sigma)} \int_{\mathbb{R}^k} \exp(-\frac{1}{2}\|y\|^2) dy$

$= \frac{1}{\sqrt{(2\pi)^k}} \left(\int_{-\infty}^{\infty} e^{-\frac{y^2}{2}} dy \right)^k$ Gauss integral

$= \frac{1}{\sqrt{(2\pi)^k}} (\sqrt{2\pi})^k = 1$

Q₂: 設 A, B 為 $n \times n$ 矩陣且 x 為 $n \times 1$ 矩陣

(a) 證明 $\frac{\partial}{\partial A} \text{trace } AB = B^T$

$$\text{tr}(AB) = \sum_{i=1}^n (AB)_{ii} = \sum_{i=1}^n \sum_{j=1}^n a_{ij} b_{ji}$$

對任一元素 A_{pq} 偏導 $\frac{\partial}{\partial A_{pq}} \text{tr}(AB) = \frac{\partial}{\partial A_{pq}} \sum_{ij} a_{ij} b_{ji} = b_{qp}$

故 $\frac{\partial}{\partial A} \text{trace } AB = B^T$

ex: $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}, B = \begin{bmatrix} p & q \\ r & s \end{bmatrix}$

$$\text{tr}(AB) = ap + br + cq + ds = f(a, b, c, d)$$

$$\frac{\partial f}{\partial a} = p \quad \frac{\partial f}{\partial b} = r \quad \frac{\partial f}{\partial c} = q \quad \frac{\partial f}{\partial d} = s$$

$$\frac{\partial f}{\partial A} = \begin{bmatrix} p & r \\ q & s \end{bmatrix} = B^T$$

(b) 證明 $x^T A x = \text{trace}(x x^T A)$

$$\text{令 } x = (x_1, \dots, x_n)^T \in M_{n \times 1}, \quad A = (A_{ij})_{n \times n} \in M_{n \times n}$$

$$1. (x^T A)_j = \sum_{i=1}^n x_i A_{ij}$$

$$x^T A x = \sum_{j=1}^n \left(\sum_{i=1}^n x_i A_{ij} \right) x_j = \sum_{i,j} x_i A_{ij} x_j$$

$$2. \text{tr}(x x^T A) = \text{tr}(A x x^T) = \sum_{i=1}^n (A x x^T)_{ii} = \sum_{i=1}^n \sum_{j=1}^n A_{ij} (x x^T)_{ji}$$

$$= \sum_{i=1}^n \sum_{j=1}^n A_{ij} x_j x_i = \sum_{i,j} x_i A_{ij} x_j$$

故 $x^T A x = \text{trace}(x x^T A)$

(c) 推導多元高斯最大概似估計值

假設有 m 個 random vector, 每個 vector size 為 p : $x^{(1)} \dots x^{(m)}$
其中每個 random vector 都可解釋成 p 個資料點. 假設
每個 $x^{(i)}$ 都是獨立分佈的多元高斯 vector: $x^{(i)} \sim N_p(\mu, \Sigma)$

$$\text{取對數: } \ell(\mu, \Sigma | x^{(i)}) = \log \prod_{i=1}^m f_{x^{(i)}}(x^{(i)} | \mu, \Sigma)$$

$$= \log \prod_{i=1}^m \frac{1}{(2\pi)^{\frac{p}{2}} |\Sigma|^{\frac{1}{2}}} \exp\left(-\frac{1}{2} (x^{(i)} - \mu)^T \Sigma^{-1} (x^{(i)} - \mu)\right)$$

$$= \sum_{i=1}^m \left(-\frac{p}{2} \log(2\pi) - \frac{1}{2} \log|\Sigma| - \frac{1}{2} (x^{(i)} - \mu)^T \Sigma^{-1} (x^{(i)} - \mu)\right)$$

$$\therefore \ell(\mu, \Sigma) = -\frac{mp}{2} \log(2\pi) - \frac{m}{2} \log|\Sigma| - \frac{1}{2} \sum_{i=1}^m (x^{(i)} - \mu)^T \Sigma^{-1} (x^{(i)} - \mu)$$

推導 $\hat{\mu}$:

$$\therefore \frac{\partial W^T A W}{\partial W} = 2 A W \quad \text{if } W \text{ doesn't depend on } A \text{ and } A \text{ is symmetric}$$

正定 Matrix

$$\therefore \frac{\partial}{\partial \mu} \ell(\mu, \Sigma) = \sum_{i=1}^m \Sigma^{-1} (x^{(i)} - \mu) = 0$$

$$0 = m\mu - \sum_{i=1}^m x^{(i)} \quad \longrightarrow \quad \hat{\mu} = \frac{1}{m} \sum_{i=1}^m x^{(i)} = \bar{x}$$

推導 $\hat{\Sigma}$:

$$\ell(\mu, \Sigma | x^{(i)}) = -\frac{mp}{2} \log(2\pi) - \frac{m}{2} \log|\Sigma| - \frac{1}{2} \sum_{i=1}^m (x^{(i)} - \mu)^T \Sigma^{-1} (x^{(i)} - \mu)$$

$$= \underbrace{C}_{\text{常數}} - \frac{m}{2} \log|\Sigma| - \frac{1}{2} \sum_{i=1}^m (x^{(i)} - \mu)^T \Sigma^{-1} (x^{(i)} - \mu)$$

$$= C + \frac{m}{2} \log|\Sigma^{-1}| - \frac{1}{2} \sum_{i=1}^m \text{tr}[\Sigma^{-1} (x^{(i)} - \mu)(x^{(i)} - \mu)^T]$$

$$\frac{\partial}{\partial \Sigma^{-1}} \ell(\mu, \Sigma | x^{(i)}) = \frac{m}{2} \Sigma - \frac{1}{2} \sum_{i=1}^m (x^{(i)} - \mu)(x^{(i)} - \mu)^T \quad \text{since } \Sigma^T = \Sigma$$

$$\longrightarrow O = m \Sigma - \sum_{i=1}^m (x^{(i)} - \mu)(x^{(i)} - \mu)^T$$

$$\longrightarrow \hat{\Sigma} = \frac{1}{m} \sum_{i=1}^m (x^{(i)} - \hat{\mu})(x^{(i)} - \hat{\mu})^T$$